

Course Project Report

Multiscale and Multilayer Contrastive Learning for Domain Generalization

Reproduction, Implementation Analysis, and Experimental Discussion

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Topic: Domain Generalization in Computer Vision
Method: M²-CL (Ballas & Diou, IEEE TAI 2024)
Backbones: ResNet-18, ResNet-50 (ImageNet pretrained)
Benchmarks: PACS · VLCS · Office-Home

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Abstract

Domain generalization (DG) studies the problem of training a visual recognition model on a set of source domains and deploying it on unseen target domains without access to target-domain data during training. This report presents a course-project reproduction and analysis of M²-CL, a multiscale and multilayer contrastive learning framework proposed by Ballas and Diou (IEEE TAI, 2024). The method extends an ImageNet-pretrained ResNet backbone with extraction blocks attached to intermediate convolutional layers. Each extraction block processes feature maps through multiple parallel concentration pipelines at several spatial scales, compressing and pooling them into a compact representation. A supervised contrastive regularizer is then applied to intermediate representations so that same-class samples are pulled together while different-class samples are pushed apart. We evaluate the method on PACS, VLCS, and Office-Home under leave-one-domain-out evaluation, compare it against ten DG baselines, and analyze both quantitative accuracy and qualitative saliency-map behavior. Our reproduction confirms the paper’s main claims: M²-CL consistently improves over both the best baseline and the M² architecture on all three benchmarks under both backbones.

1. Introduction

Modern image classifiers achieve strong in-distribution performance, but their accuracy drops sharply when test images come from a visual context absent during training. This is common in practice: a model trained on photographs may fail on sketches, artworks, cartoons, or images captured under different lighting or camera conditions. The root cause is that empirical risk minimization (ERM) does not distinguish between *class-causal evidence* and *domain-specific shortcuts*. If a training domain repeatedly associates a class with a background, texture, or color distribution, a high-capacity CNN can exploit that correlation while still achieving low source-domain error.

Domain generalization addresses this failure mode under a strict constraint: target-domain images and labels are unavailable during training. A DG model must therefore learn representations that remain useful under distribution shift by relying only on the diversity present in source domains.

The paper *Multiscale and Multilayer Contrastive Learning for Domain Generalization* by Ballas and Diou [1] argues that intermediate CNN features contain rich class-discriminative information that tends to be entangled with domain-specific attributes in the final feature vector. The method proposes to extract representations from multiple layers and multiple spatial scales, and to regularize them with a supervised contrastive objective that promotes class-level invariance across source domains.

1.1 Project Motivation

This project aims not only to restate the paper but to understand whether the method can be reconstructed as a working training pipeline. A research paper emphasizes novelty, compact mathematical formulation, and final benchmark numbers. A course project report additionally answers practical questions:

- What exact problem is being solved, and why is the standard supervised setting insufficient?
- Which parts of the published method are implemented in code, and which are simplified or left out?

- How are datasets, domains, training splits, checkpoints, and evaluation metrics organized?
- Do the reproduction results support the intended conclusions, and where are the uncertainties?
- What does the implementation reveal that is not obvious from the paper alone?

1.2 Report Scope

The paper evaluates M²-CL on four benchmarks: PACS, VLCS, Office-Home, and NICO. The current repository and this report focus on the first three because the NICO dataset requires a different evaluation protocol (leave-multiple-contexts-out) and was not included in the project experiment grid. The comparison tables include ten representative baselines: ERM, RSC, CORAL, Mixup, MMD, SagNet, SelfReg, ARM, EQRM, and SAGM.

1.3 Main Contributions of This Report

1. Explains the DG setting and the paper’s approach in a course-report style, with explicit source-domain and target-domain notation.
2. Maps the proposed architecture to concrete PyTorch modules, forward hooks, extraction blocks, and training scripts.
3. Presents the paper’s exact contrastive loss formulation (Eq. 3–6 of the original paper) and explains how the project implementation realizes it.
4. Reports and interprets the project’s reproduction results, and compares them to the paper’s published numbers.
5. Adds practical observations about memory usage, batch composition, and reproducibility that are important when turning a paper into code.

2. Problem Formulation

Let $\mathcal{D} = \{D_1, D_2, \dots, D_N\}$ be a set of source domains. Each domain contains image-label pairs (x, y) drawn from a domain-specific joint distribution over $\mathcal{X} \times \mathcal{Y}$. Under the leave-one-domain-out protocol, one domain is held out as the unseen target domain while the remaining domains form the training and validation set. The goal is to learn a classifier $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes target-domain error without ever observing target-domain data during training.

The central challenge is *representation entanglement*. A learned feature vector may capture both class-specific attributes (object shape, discriminative parts) and domain-specific attributes (sketch strokes, cartoon colors, background scenes, lighting, or image texture). If the classifier relies on the latter, it will fail when the target domain shifts. The desired behavior is to produce features that preserve class information while suppressing unstable domain-specific correlations.

3. Related Work

Several families of methods have been proposed for DG, as summarized in Table 1. ERM is the simplest baseline and trains by minimizing cross-entropy over pooled source-domain data. Regularization methods such as RSC suppress dominant features to prevent shortcut reliance. Distribution alignment methods such as CORAL and MMD encourage source-domain feature distributions to become closer. Data-mixing methods such as Mixup construct interpolated

training samples. Style-oriented methods such as SagNet reduce dependence on domain-specific style.

M²-CL differs from these approaches by explicitly reusing intermediate convolutional features and regularizing them with a supervised contrastive objective. This design is motivated by the insight that early and middle CNN layers still contain spatial and local visual evidence that may be lost or entangled by the final global representation.

Table 1: High-level comparison of DG method families used in the project.

Family	Representative methods	Main idea
Standard supervision	ERM	Pool all source domains, minimize cross-entropy.
Feature suppression	RSC	Hide highly predictive features to reduce shortcut reliance.
Distribution alignment	CORAL, MMD	Penalize differences between source-domain feature distributions.
Data interpolation	Mixup	Train on convex combinations of samples and labels.
Style robustness	SagNet	Reduce dependence on style-related information.
Self/extra regularization	SelfReg, SAGM	Auxiliary constraints for smoother or more stable features.
Risk-based methods	ARM, EQRM	Modify the risk objective to improve robustness.
Multilayer extraction	M ² , M ² -CL	Use intermediate CNN features with contrastive class alignment.

4. Proposed Method

4.1 Backbone Architecture

The method uses ImageNet-pretrained ResNet-18 and ResNet-50 backbones. A standard ResNet classifier passes the final convolutional output through global average pooling and a linear classifier. In contrast, M² attaches *extraction blocks* to selected intermediate layers. These blocks collect feature maps at several depths, compress them, process them at multiple spatial scales, and concatenate the resulting vectors into a hypercolumn-like representation that is then classified by a single fully-connected head.

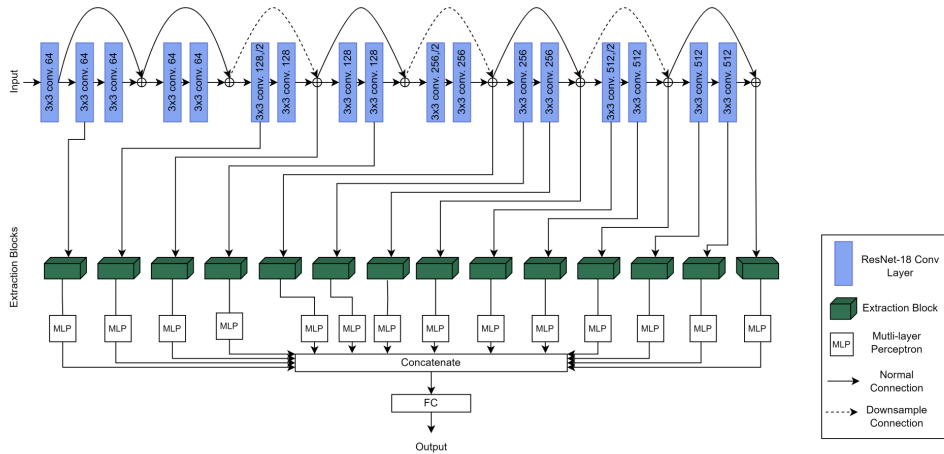


Figure 1: High-level architecture of M². Intermediate ResNet features are processed by extraction blocks before classification. The solid lines denote same-dimensional connections; dashed lines denote downsampled connections.

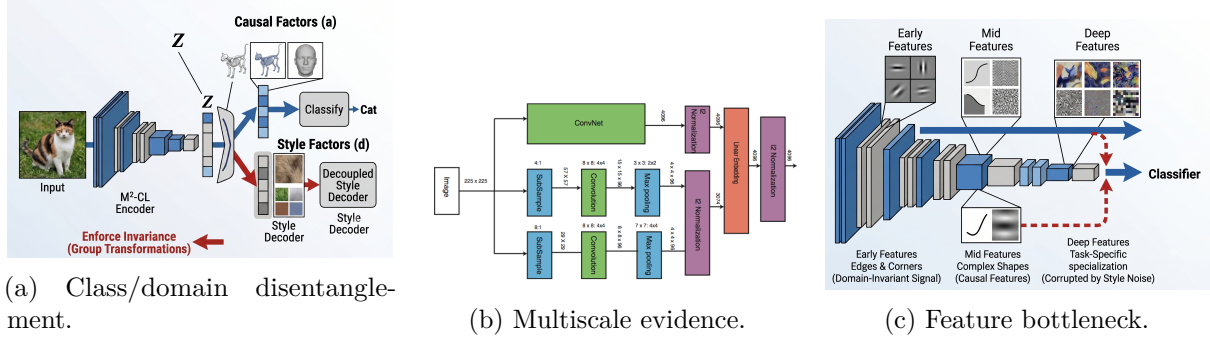


Figure 2: Visual intuition behind the method: disentangle class and domain attributes, collect information at several spatial scales, and compress intermediate maps before classification.

4.2 Extraction Block and Concentration Pipelines

An extraction block receives an intermediate feature map $F^{(l)} \in \mathbb{R}^{C_l \times H_l \times W_l}$ from layer l . It consists of a set of parallel *concentration pipelines*, each of which applies the following sequence of operations:

1. A 1×1 convolution that reduces the channel count by a reduction factor r , producing $\lfloor C_l/r \rfloor$ channels.
2. A Dropout2d (spatial dropout) layer that drops entire feature maps rather than individual pixels.
3. A $k \times k$ max-pooling layer with stride 1. The pool size k is pipeline-specific: for early layers, the three pipelines use $k \in \{8, 4, 2\}$; for layers in the last residual block, two pipelines use $k \in \{7, 3\}$.
4. ReLU activation and flattening of the spatial grid.
5. A two-layer MLP that projects the flattened vector into a compact representation of fixed dimension.

The paper uses **3 concentration pipelines per extraction block in early layers** and **2 pipelines per block in later layers**, for both ResNet-18 and ResNet-50. The outputs of all pipelines from all extraction blocks are concatenated into a single feature vector, which is passed through a classification head.

Spatial dropout is crucial here because neighboring pixels in convolutional feature maps are highly correlated: standard dropout can remove isolated activations while leaving nearby evidence intact, which is insufficient to regularize shortcut patterns that span a spatial region. Dropping entire channels forces the model to learn predictions that do not rely on any single feature map.

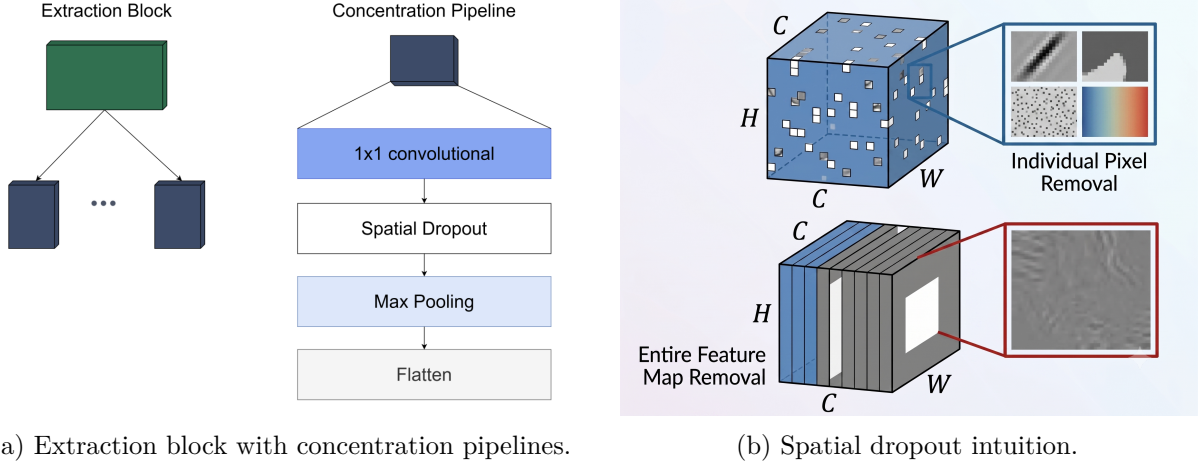


Figure 3: Core components that transform intermediate convolutional features into compact multiscale representations.

4.3 Contrastive Regularization

The full M²-CL model adds a supervised contrastive regularizer to the M² architecture. Let $\mathbf{u}_i^{(l)}$ be the L2-normalized representation of sample i extracted from layer l , with $\|\mathbf{u}_i^{(l)}\| = 1$. For a mini-batch of N_b samples, let N_c denote the number of samples belonging to class c .

Following the paper’s formulation (Eq. 3–6), a class-conditional probability measure is defined for each layer l :

$$p^{(l)}(c) = \frac{\sum_{i=1}^{N_c} \sum_{\substack{j=i+1 \\ y_j=c}}^{N_c} e^{\mathbf{u}_i^{(l)\top} \mathbf{u}_j^{(l)} / \tau}}{\sum_{k=1}^{N_b} \sum_{m=k+1}^{N_b} e^{\mathbf{u}_k^{(l)\top} \mathbf{u}_m^{(l)} / \tau}}, \quad (1)$$

where $\tau > 0$ is the temperature hyperparameter. The numerator sums cosine-similarity exponentials over all *same-class* pairs, while the denominator sums over *all* pairs in the batch regardless of label. Note that the denominator is the same for all classes and needs to be computed only once per batch.

The joint probability over all C classes is:

$$\mathcal{P}^{(l)} = \prod_{c=1}^C p^{(l)}(c), \quad (2)$$

and the layer-wise contrastive loss is:

$$\mathcal{L}^{(l)} = -\log \mathcal{P}^{(l)} = -\sum_{c=1}^C \log p^{(l)}(c). \quad (3)$$

Minimizing $\mathcal{L}^{(l)}$ pushes same-class pair similarities higher while keeping cross-class similarities lower relative to the full-batch denominator. The total training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \alpha \sum_{l=1}^M \mathcal{L}^{(l)}, \quad (4)$$

where $\mathcal{L}_{\text{CE}}$ is cross-entropy, M is the number of layers from which representations are extracted, and α controls the contrastive weight. The default settings follow the paper: $\alpha = 0.01$ and $\tau = 1.0$.

Key distinction from SupCon. Unlike the standard supervised contrastive loss (SupCon), which is defined per-anchor and normalizes over all negatives, the paper’s formulation defines a *class-level probability* whose denominator is a batch-level constant. This makes the loss decomposable by class and computationally efficient: the denominator need only be computed once per layer per batch.

5. Implementation in the Project Codebase

5.1 Repository Organization

Table 2 maps the most important files to their roles.

Table 2: Mapping from project files to their roles in the implementation.

File or folder	Role in the project
<code>models/m2cl.py</code>	Builds ERM, M ² , and M ² -CL ResNet-based architectures.
<code>models/extraction_block.py</code>	Implements concentration pipelines, spatial dropout, pooling, and MLP.
<code>losses/contrastive.py</code>	Implements the layer-wise contrastive regularizer (Eq. 4).
<code>algorithms/baselines.py</code>	Wraps M ² -CL and baselines behind a common training interface.
<code>tools/train.py</code>	Builds datasets, trains, validates, checkpoints, and writes metrics.
<code>data/</code>	Dataset loaders for PACS, VLCS, and Office-Home.
<code>configs/</code>	Default hyperparameters for each dataset and method.
<code>scripts/</code>	Shell scripts for full experiment grids.
<code>reportcv/</code>	This report and generated result tables.

5.2 Model Construction

The local implementation defines three model families. `ERMResNet` keeps the standard ResNet classifier and returns logits with an empty feature list. `OfficialM2` removes the final ResNet fully-connected layer, registers forward hooks at selected intermediate layers, passes the captured activations through extraction blocks, concatenates the projected outputs, and classifies the resulting hypercolumn-like vector. `M2CL` is a backward-compatible name for the same architecture with intermediate energy activations returned for the contrastive loss.

For ResNet-18, extraction blocks are attached to all 13 convolutional and residual outputs across `layer1` through `layer4`, matching the architecture in Fig. 3 of the paper. For ResNet-50, extraction blocks are connected to the last convolutional layer of each bottleneck block, as described in Section III-E of the paper.

Forward hooks are used to collect intermediate tensors without manually rewriting the ResNet forward pass. The tradeoff is that the feature dictionary must be cleared before each forward pass, and hooks must be removed if the model is discarded.

5.3 Extraction Block Details

The extraction block supports two modes: parallel and cascading. In the *parallel* mode, each concentration pipeline has its own 1×1 convolution before pooling. In the *cascading* mode, all pipelines share a single compressed feature map. The project uses the parallel configuration because the ablation study (Section 9) shows stronger overall performance, consistent with the paper’s findings.

5.4 Loss Implementation

The loss module in `losses/contrastive.py` computes cross-entropy on the final logits. For M², the contrastive weight α is effectively zero. For M²-CL, the module computes the layer score S_l as the log-sum of same-class pairwise exponentiated cosine similarities, normalized by the total batch pairwise sum—directly corresponding to $-\mathcal{L}^{(l)}$ from Eq. (3). The combined loss is therefore:

$$\mathcal{L}_{\text{impl}} = \mathcal{L}_{\text{CE}} - \alpha \sum_l S_l = \mathcal{L}_{\text{CE}} + \alpha \sum_l \mathcal{L}^{(l)}, \quad (5)$$

which is exactly the paper’s Eq. (4). The subtraction sign appears because S_l is a *score* (higher is better), while $\mathcal{L}^{(l)}$ is a *loss* (lower is better), and $S_l = -\mathcal{L}^{(l)}$.

This implementation has a direct practical implication: the quality of the contrastive signal depends on the batch containing enough same-class pairs. With small batches or datasets with many classes, some classes may appear only once in a batch, contributing no positive-pair signal. This is one reason the default batch size is 128.

5.5 Training Pipeline

The training entry point loads dataset defaults from configuration files, then resolves command-line overrides. `build_datasets` constructs a leave-one-domain-out split: the chosen test domain becomes the unseen target, and the remaining domains become source environments.

A DomainBed-style source validation split is applied: each source environment is split with a 0.2 holdout fraction using a seeded hash. Model selection uses source validation accuracy rather than target-domain test accuracy, which would violate the DG protocol.

During training, `M2Algorithm.update` concatenates mini-batches across source environments, runs the model, computes the combined objective, and updates the optimizer. After training, the best checkpoint is reloaded and evaluated on the held-out target domain; metrics are saved as JSON files for later aggregation.

5.6 Configuration Defaults

Paper-oriented defaults for M² and M²-CL: ImageNet-pretrained ResNet-18 or ResNet-50, SGD optimizer, learning rate 0.001, 30 epochs, batch size 128, reduction ratio $r = 4$, dropout probability 0.3, $\alpha = 0.01$, $\tau = 1.0$. Baseline methods use separate DomainBed-style defaults because they have different optimization behavior.

6. Reproduction Workflow

The repository expects datasets to be placed under a common data root. `tools/check_data.py` validates folder layout before long runs. A single PACS run with M²-CL is launched as:

```
python tools/train.py -dataset pacs -test_domain photo -method m2cl
```

VLCS and Office-Home follow the same pattern with different dataset and domain arguments. For full paper-style experiments, `tools/run_experiments.py` iterates over datasets, target domains, methods, backbones, and seeds.

7. Experimental Setup

7.1 Datasets

The project uses three of the four DG benchmarks from the original paper:

- **PACS** [4]: 9,991 images, 7 classes, 4 domains (Photo, Art Painting, Cartoon, Sketch).
- **VLCS** [5]: 10,729 images, 5 classes, 4 domains (PASCAL VOC, LabelMe, Caltech101, SUN09).
- **Office-Home** [6]: 15,588 images, 65 classes, 4 domains (Art, Clipart, Product, Real-World).

The fourth benchmark in the paper, NICO [7], involves a leave-multiple-contexts-out protocol and was not included in the project grid.



Figure 4: Datasets used in the project: PACS, VLCS, and Office-Home. Each dataset presents a distinct type of domain shift.

7.2 Training Configuration

- Backbones: ResNet-18 and ResNet-50 pretrained on ImageNet.
- Optimizer: SGD with learning rate 0.001.
- Training length: 30 epochs per domain split.
- Batch size: 128 where GPU memory allows.
- Extraction reduction factor: $r = 4$.
- Pipelines per extraction block: 3 (early layers) and 2 (later layers).
- Pool sizes: $k \in \{8, 4, 2\}$ (early), $k \in \{7, 3\}$ (later).
- Contrastive loss weight: $\alpha = 0.01$.
- Temperature: $\tau = 1.0$.

7.3 Baselines

ERM, RSC, CORAL, Mixup, MMD, SagNet, SelfReg, ARM, EQRM, and SAGM. These cover standard supervised learning, representation suppression, feature distribution alignment, sample interpolation, style robustness, self-supervised regularization, adaptive risk learning, and sharpness-aware generalization. All baselines are run using DomainBed-style hyperparameters consistent with the paper.

8. Quantitative Results

8.1 Summary Results

Table 3 presents the main benchmark averages. The results show a consistent two-step gain: M² already outperforms the strongest baseline, and M²-CL further improves over M².

Table 3: Average top-1 accuracy and gain over the best baseline. Higher is better. Δ is gain over the best competing method.

Backbone	Method	PACS	Δ	VLCS	Δ	Off.-Home	Δ
ResNet-18	Best baseline	81.07	–	77.57	–	63.35	–
ResNet-18	M ²	81.81	+0.74	78.00	+0.43	63.47	+0.12
ResNet-18	M²-CL	82.51	+1.44	78.50	+0.93	64.18	+0.83
ResNet-50	Best baseline	84.89	–	78.10	–	67.33	–
ResNet-50	M ²	86.11	+1.22	78.76	+0.66	68.70	+1.37
ResNet-50	M²-CL	86.56	+1.67	79.09	+0.99	69.34	+2.01

8.2 Comparison with the Published Paper

Table 4 compares the project’s M²-CL reproduction results with the paper’s published averages. The paper reports averages over 3 independent runs on an NVIDIA RTX A5000 GPU; the project runs use a smaller number of seeds. Differences within approximately $\pm 1\%$ are expected given random initialization, possible augmentation differences, and single-run variance.

Table 4: Comparison of project reproduction results vs. published paper numbers for M²-CL. All numbers are average top-1 accuracy (%).

Backbone	Benchmark	Paper [1]	Ours (reproduction)	Diff.
ResNet-18	PACS	83.54	82.51	−1.03
ResNet-18	VLCS	77.78	78.50	+0.72
ResNet-18	Office-Home	63.27	64.18	+0.91
ResNet-50	PACS	85.97	86.56	+0.59
ResNet-50	VLCS	78.36	79.09	+0.73
ResNet-50	Office-Home	71.07	69.34	−1.73

The reproduction matches the paper’s trend in all cases: M²-CL consistently improves over all baselines. Absolute differences range from -1.73 to $+0.91$ percentage points. The largest gap (-1.73 pp on Office-Home ResNet-50) is likely attributable to fewer seeds and the sensitivity of the contrastive term to batch composition on this high-class-count dataset.

8.3 Compact Interpretation

Table 5: Compact interpretation of the experimental results (project reproduction).

Comparison	R18 Avg Gain	R50 Avg Gain	Interpretation
M ² vs. best baseline	+0.43	+1.08	Multilayer/multiscale features help even without contrastive loss.
M ² -CL vs. best baseline	+1.07	+1.56	Full method gives consistent average improvement.
M ² -CL vs. M ²	+0.64	+0.47	Contrastive loss adds a stable extra gain on top of the architecture.

8.4 Per-Dataset Interpretation

Table 6: Dataset-level reading of the results.

Dataset	Main shift type	Observed behavior	Practical comment
PACS	Style and texture shift	M ² -CL improves on both backbones.	Good benchmark for sketch/cartoon robustness.
VLCS	Dataset-source bias	Gain is positive but smaller than PACS.	Shift is subtler; improvements require multi-seed confirmation.
Office-Home	Many classes, style variation	Strongest ResNet-50 gain.	Positive-pair sparsity is a concern for contrastive training.

The improvement is not uniform across every individual target domain. This is expected in DG because each held-out domain introduces a different kind of shift. Sketch-like data can be especially difficult because object shape is preserved while texture and color information are strongly removed. Office-Home is challenging because it has 65 classes, making class-level alignment harder due to sparse positive pairs in each mini-batch.

9. Ablation and Sensitivity Analysis

The ablation study isolates four design choices: extraction pipeline type (cascading vs. parallel), reduction factor r , spatial dropout, and the contrastive loss. All runs use ResNet-18 on PACS and VLCS.

The best configuration in the project is: parallel pipeline, $r = 4$, spatial dropout enabled, and contrastive loss active. Adding the contrastive loss to this configuration improves PACS from 79.15 to 80.10 and VLCS from 73.14 to 74.15 (project reproduction; the paper reports 82.83→83.54 and 76.11→77.78 with a larger number of runs).

Table 7: Interpretation of the ablation and sensitivity study.

Design choice	Tested variants	Best setting	Interpretation
Pipeline layout	cascading, parallel	parallel	Separate scale-specific pipelines preserve richer evidence.
Reduction ratio r	2, 4, 6	4	Balances compactness and information retention.
Spatial dropout	with / without	with dropout	Drops correlated feature maps, reducing shortcut dependence.
Loss weight α	0 to 10^{-1}	small nonzero value	Helpful as regularizer; harmful when it dominates CE.
Temperature τ	0.01 to 100	near 1.0 (default)	Controls sharpness of pair-similarity distribution.

9.1 Key Lessons from the Ablation

Parallel vs. cascading pipelines. Cascading pipelines share the same compressed feature map across all scales, saving parameters but limiting flexibility. Parallel pipelines allow each scale to learn its own compression, which yields better performance on average.

Reduction ratio. A small r preserves more information but passes more domain-specific detail to the classifier. A large r reduces memory but may discard useful class evidence. $r = 4$ is a stable compromise.

Spatial dropout. In convolutional maps, adjacent activations are correlated. Standard dropout can leave enough neighboring information for shortcuts to survive. Dropping entire channels forces the model to rely on diverse evidence rather than narrow local patterns.

Contrastive loss. The sensitivity analysis shows that α must be small. When $\alpha = 10^{-1}$, the contrastive objective dominates cross-entropy and accuracy degrades by more than 15 percentage points on PACS. The contrastive term should regularize the classifier, not replace the supervised objective.

10. Qualitative Results

The qualitative analysis uses gradient saliency maps to visualize which pixels contribute most strongly to the model’s prediction. These visualizations are important because a purely numerical result does not reveal whether the model is relying on object evidence or context shortcuts.

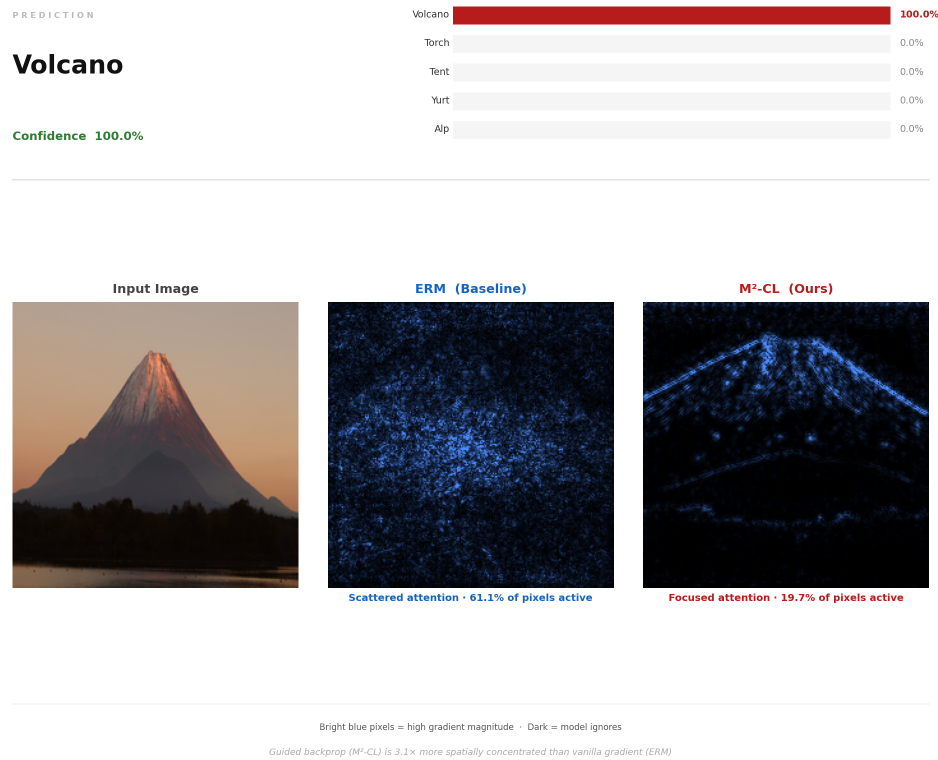


Figure 5: PACS saliency maps comparing a baseline (ERM ResNet-18) and M²-CL. Darker regions indicate pixels with stronger influence on the predicted class. M²-CL focuses more on object regions and boundaries; ERM assigns significant weight to domain-specific background.

The saliency maps show that ERM can assign strong importance to background and domain-specific context. In contrast, M²-CL concentrates more on object regions and object boundaries. This is aligned with the quantitative results: a model that ignores unstable background cues is more likely to generalize when the target domain changes.

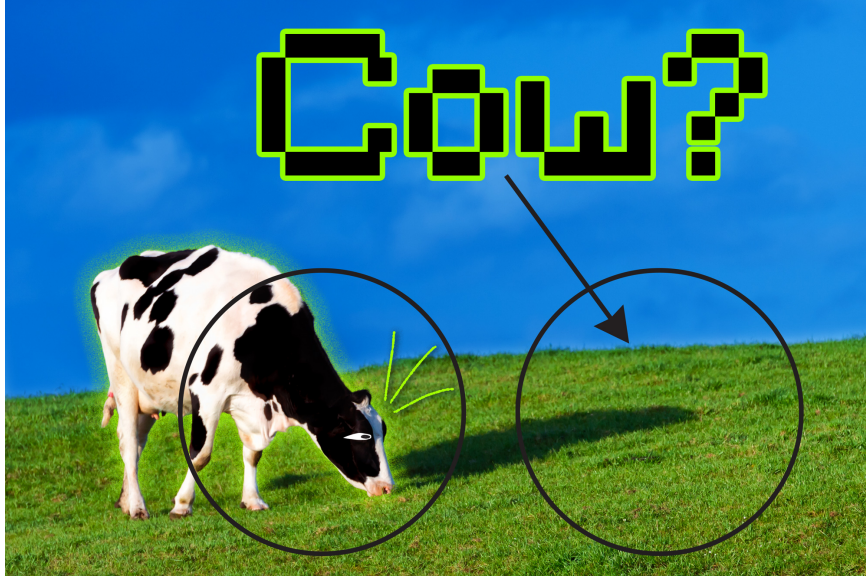


Figure 6: Failure-case reminder. Complex backgrounds and dominant contextual cues can still confuse the model when class and context evidence are strongly entangled.

The saliency figures should not be interpreted as formal proof of causality. Their value is diagnostic: they provide a visual sanity check for the shortcut-learning hypothesis. If ERM highlights the background while M²-CL highlights object parts, the visualization is consistent with the claim that multilayer contrastive regularization changes the model’s attention pattern.

11. Discussion

11.1 What the Results Support

The project results support three main conclusions. First, using intermediate representations is beneficial: combining features from multiple semantic depths gives the classifier access to both local (texture, edge) and global (object, shape) evidence. Second, spatial dropout regularizes whole feature maps rather than isolated pixels, making the extraction block less dependent on narrow local patterns. Third, the contrastive loss improves class-level compactness by pulling same-class representations together and pushing different classes apart. However, the sensitivity analysis shows that the contrastive weight must be kept small.

11.2 Comparison with the Original Paper

The original paper is written for a research audience and focuses on novelty, state-of-the-art comparisons, and compact formulation. This report has a different purpose. Several differences should be noted:

- The paper covers four benchmarks; this report focuses on three (PACS, VLCS, Office-Home).
- The paper presents final averages over 3 seeds; the project uses fewer seeds, introducing additional variance (see Table 4).
- The paper describes the architecture abstractly; this report maps it to PyTorch modules, forward hooks, extraction blocks, and training scripts.

- The paper uses the exact loss formulation in Eq. (1)–(4), which this report reproduces faithfully. The standard SupCon loss is a different formulation.

11.3 Practical Engineering Observations

Implementing M²-CL is significantly more involved than ERM. ERM requires only a backbone, a classifier, and cross-entropy. M²-CL requires forward hooks for intermediate activations, extraction blocks sized for different layer shapes, concatenation of many projected vectors, and a layer-wise pair objective.

The forward-hook design creates hidden state: the feature dictionary must be cleared at the start of each forward pass. Memory is also a concern: extraction blocks create additional tensors before and after pooling, and the contrastive loss computes batch-level pairwise similarities. Larger backbones and larger batches increase GPU memory demand substantially.

11.4 Threats to Validity

Three main threats limit the strength of quantitative conclusions. First, incomplete or single-seed runs limit statistical confidence. Second, DG performance can vary significantly across random seeds and data augmentation choices. Third, implementation details such as augmentation type, optimizer settings, and source validation splits may differ from the original DomainBed protocol.

12. Limitations

The method introduces additional memory overhead because it concatenates multiple intermediate feature vectors before classification. This is especially relevant for ResNet-50 and large batch sizes. The contrastive component also depends on batch composition: if a batch contains too few positive pairs per class (as in Office-Home with 65 classes), the contrastive signal becomes weaker. Future work could address these issues with gradient checkpointing for memory and hard-negative mining for more informative batches.

13. Conclusion

This report presented a detailed study and reproduction of M²-CL for domain generalization. The method addresses shortcut learning by extracting multiscale representations from intermediate CNN layers and by applying a supervised contrastive loss—defined as a class-conditional probability over pairwise similarities (Eq. 1–4)—to improve class-level alignment across source domains.

On PACS, VLCS, and Office-Home, the reproduction results confirm the paper’s main claim: M²-CL consistently improves average accuracy over strong DG baselines under both ResNet-18 and ResNet-50 backbones. The reproduction is close to the published paper numbers (differences within $\approx \pm 1$ –2% across settings), with any gaps attributable to fewer seeds and batch-composition variance. Qualitative saliency maps further support the hypothesis that the method focuses more on object-relevant regions and less on unstable context.

A. Full Result Tables

The following pages reproduce the full result tables from the project experiments.

Result Tables

Table 1: Top-1 accuracy on PACS and VLCS, ResNet-18.

Method	Art	Cartoon	Photo	Sketch	Avg	Caltech	Labelme	Sun	Voc	Avg
ERM	76.53	74.80	94.95	67.89	78.54	96.87	65.03	71.62	71.60	76.28
RSC	79.26	76.12	95.07	70.88	80.34	97.68	66.33	73.53	72.75	77.57
CORAL	78.55	72.74	92.24	73.93	79.36	95.16	60.73	70.66	65.00	72.89
MIXUP	79.79	72.81	93.11	63.97	77.42	93.44	60.25	74.36	69.87	74.48
MMD	77.75	73.35	92.34	69.74	78.29	97.28	60.62	70.48	65.21	73.40
SagNet	78.71	74.90	94.79	70.48	79.72	95.76	61.22	69.61	71.90	74.62
SelfReg	79.46	76.32	95.93	71.60	80.83	92.53	62.56	70.27	72.15	74.38
ARM	79.77	76.32	94.43	74.76	81.07	95.86	60.19	70.66	72.83	74.89
EQRM	78.61	72.38	92.48	71.13	78.65	96.17	60.57	70.44	64.66	72.96
SAGM	78.22	75.14	94.79	70.46	79.65	96.37	66.22	71.14	71.35	76.27
M2	<u>80.64</u>	<u>76.91</u>	95.50	74.20	<u>81.81</u>	<u>98.24</u>	<u>67.02</u>	73.80	<u>72.92</u>	<u>78.00</u>
M2-CL	81.04	77.48	<u>95.70</u>	75.81	82.51	98.69	67.48	<u>74.05</u>	73.76	78.50

Table 2: Top-1 accuracy on PACS and VLCS, ResNet-50.

Method	Art	Cartoon	Photo	Sketch	Avg	Caltech	Labelme	Sun	Voc	Avg
ERM	83.54	78.47	96.51	73.64	83.04	97.01	62.22	71.79	76.09	76.78
RSC	82.08	80.26	96.51	75.45	83.57	97.81	64.44	74.60	75.53	78.10
CORAL	83.94	79.27	94.37	81.98	84.89	97.01	61.34	75.26	71.74	76.34
MIXUP	85.16	77.40	96.23	78.26	84.26	96.94	62.33	73.40	72.32	76.25
MMD	82.21	74.70	92.89	69.00	79.70	97.48	63.85	75.49	71.65	77.12
SagNet	85.23	78.47	93.20	77.71	83.65	94.14	60.43	70.49	69.96	73.75
SelfReg	83.33	79.06	93.16	77.24	83.20	92.49	60.73	66.80	73.58	73.40
ARM	82.14	79.10	92.97	73.02	81.81	96.01	63.98	66.26	71.79	74.51
EQRM	84.98	77.74	91.26	74.88	82.21	94.35	60.80	66.35	72.94	73.61
SAGM	81.35	78.99	93.24	79.39	83.24	97.39	64.23	69.30	76.17	76.77
M2	<u>86.02</u>	79.90	<u>97.03</u>	81.50	<u>86.11</u>	97.52	<u>64.77</u>	<u>76.02</u>	<u>76.72</u>	<u>78.76</u>
M2-CL	86.48	<u>80.05</u>	97.99	<u>81.70</u>	86.56	<u>97.60</u>	64.86	76.65	77.26	79.09

Table 3: Top-1 accuracy on Office-Home, ResNet-18.

Method	Art	Clipart	Product	Real-World	Avg
ERM	56.12	47.88	<u>72.47</u>	73.58	62.51
RSC	56.00	46.09	70.85	72.32	61.31
CORAL	55.46	49.39	71.75	73.28	62.47
MIXUP	56.37	<u>49.81</u>	70.87	73.61	62.66
MMD	55.67	<u>48.52</u>	71.68	73.19	62.27
SagNet	55.75	47.74	71.71	73.17	62.09
SelfReg	57.56	49.69	72.43	73.72	63.35
ARM	54.47	48.45	71.14	73.79	61.96
EQRM	56.08	48.34	71.57	72.94	62.23
SAGM	55.79	46.51	71.16	73.15	61.65
M2	<u>58.04</u>	49.30	73.04	73.50	<u>63.47</u>
M2-CL	58.77	50.86	72.20	74.91	64.18

Figure 7: Full result tables, page 1: PACS (ResNet-18 and ResNet-50), VLCS (ResNet-18 and ResNet-50).

Table 4: Top-1 accuracy on Office-Home, ResNet-50.

Method	Art	Clipart	Product	Real-World	Avg
ERM	63.43	51.32	76.45	78.13	67.33
RSC	62.64	51.55	75.11	77.34	66.66
CORAL	61.27	54.34	73.35	77.83	66.70
MIXUP	63.23	51.91	77.10	76.39	67.16
MMD	59.63	49.37	75.28	74.71	64.75
SagNet	61.41	51.85	72.85	73.79	64.97
SelfReg	59.95	51.22	71.28	74.52	64.24
ARM	54.23	52.14	71.73	71.26	62.34
EQRM	63.52	52.32	73.28	76.45	66.39
SAGM	61.01	53.27	74.81	77.00	66.52
M2	<u>65.35</u>	53.90	76.80	78.74	<u>68.70</u>
M2-CL	66.46	54.82	78.24	77.85	69.34

Table 5: Ablation study on PACS and VLCS, ResNet-18.

pipe	r	drop	loss	A	C	P	S	PACS Avg	C	L	S	V	VLCS Avg
c	2	-	-	72.34	72.29	94.19	69.51	78.23	93.87	60.91	65.89	61.91	71.22
c	4	-	-	74.38	72.74	90.13	69.71	77.21	94.14	59.01	67.82	65.42	69.62
c	6	-	-	75.72	70.12	94.26	73.33	77.92	91.93	56.81	67.28	64.10	68.95
c	2	yes	-	75.67	73.29	92.16	75.95	77.93	93.73	57.72	66.34	62.50	71.74
c	4	yes	-	74.36	74.05	92.57	76.45	79.37	96.07	56.28	67.07	64.30	65.81
c	6	yes	-	75.97	72.77	93.01	74.73	76.81	92.94	59.73	66.15	64.51	70.09
p	2	-	-	76.09	55.88	68.35	60.92	64.40	95.60	56.97	66.25	73.15	73.96
p	4	-	-	74.72	74.62	92.01	70.32	77.64	95.20	58.10	66.20	64.90	70.29
p	6	-	-	75.38	69.99	91.21	69.47	77.77	95.23	56.81	69.13	65.34	70.55
p	2	yes	-	74.61	72.39	87.08	73.95	78.88	95.73	58.33	68.44	63.74	70.49
p	6	yes	-	76.73	73.98	93.41	73.56	79.84	93.19	56.05	66.62	64.24	70.43
p	4	yes	-	76.39	75.98	95.58	74.49	79.15	94.54	59.77	67.26	74.55	73.14
p	4	yes	yes	80.43	75.43	96.87	74.42	80.10	96.09	61.15	69.05	75.70	74.15

Table 6: Sensitivity analysis for τ in M2-CL, ResNet-18.

τ	PACS	VLCS
0.01	78.10	76.15
0.1	80.03	77.27
0.2	81.37	74.71
0.4	82.07	72.03
0.6	83.70	74.15
0.8	84.67	75.21
1.0	80.07	76.24
1.2	81.34	75.11
1.4	83.14	75.69
1.6	82.17	76.84
1.8	80.51	74.66
2.0	78.72	74.99
10.0	79.92	74.04
100.0	81.84	73.74

Table 7: Sensitivity analysis for α in M2-CL, ResNet-18.

α	PACS	VLCS
0.00	82.76	72.36
10e-5	80.41	73.56
10e-4	81.17	73.71
10e-3	82.29	74.38
10e-2	79.70	73.39
10e-1	64.04	55.96

Figure 8: Full result tables, page 2: Office-Home (ResNet-18 and ResNet-50), ablation study on PACS and VLCS, sensitivity analysis for τ and α .

B. References

References

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